

Insertion, Evasion, and Denial of Service: Eluding Network Intrusion Detection

Problem Statement

The paper addresses the weaknesses of network intrusion detection systems (NIDS) in handling evasion techniques. Specifically, it examines how attackers can manipulate network traffic to evade detection, making it difficult for NIDS to identify malicious activities accurately. The authors focus on techniques such as insertion, evasion, and denial-of-service (DoS) attacks that exploit protocol ambiguities to mislead intrusion detection mechanisms and point out the systemic flaws in Intrusion Detection Systems

Assumptions

One hidden assumption of this paper is that in order to replicate or develop upon this work one must have good knowledge of the CASL tool.

Pros

- The paper provides a breakdown of various evasion techniques and their impact on NIDS. It includes examples and demonstrating how these evasions can bypass detection.
- I personally liked that the authors highlighted fundamental issues in packet reassembly, fragmentation, and session reconstruction.

Cons

- One big issue for me was that while the paper discusses attack techniques in depth, it does not propose fully developed solutions to counteract them.
- Assumption of Static NIDS – The research focuses on weaknesses in traditional NIDS and does not account for advancements in modern, adaptive intrusion detection systems.
- Although the paper is extremely detailed the authors have not released their code implementations of the attacks.
- The CASL tool I found very less information about it online, maybe its not an open source tool so that will also be one of the cons of the paper as its not open source so hard for researchers to get.

Bro: A System for Detecting Network Intruders in Real-Time

Problem Statement

This one shows the need for a high-speed, real-time intrusion detection system (IDS) capable of passively monitoring network traffic. It presents Bro, a system designed to detect network intrusions efficiently while maintaining a clear separation between detection mechanisms and security policy. The research focuses on building a standalone network monitor that can be deployed without modifying host systems.

Assumptions

The only big assumption is already state clearly in the paper:

We must assume that attackers will (eventually) have full knowledge of the techniques used by the monitor, and access to its source code, and will use this knowledge in attempts to subvert or overwhelm the monitor so that it fails to detect the attacker's break-in activity. This assumption significantly complicates the design of the monitor; but failing to address it is to build a house of cards.

Pros

- The most important + point of this paper is that they have given this whole system publicly in the code form so people can test, analyze and modify it.
- Separation of Mechanism and Policy by allowing site-specific security policies through a scripting language.
- Its flexible because new detection techniques and attack patterns can be integrated easily.

Cons

- Upon looking at the complexity of this System I presume it must consume lot of resources.
- Another big problem of this paper is that they did the minimal by only making a detection sytem and not a system which takes some action regarding the attack.